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Design of an Intelligent Navigation System Integrating Reinforcement Learning and Computer Vision Algorithms

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Abstract

Traditional static guidance systems have problems such as poor interactivity and low path planning efficiency. This paper designs an intelligent guidance system to achieve efficient, accurate and personalized navigation services. This paper constructs an intelligent guidance system that integrates reinforcement learning and computer vision algorithms, and adopts a multi-level architecture: the perception layer collects environmental data, the data processing layer uses YOLO and semantic segmentation to extract features, the decision layer uses deep Q network (DQN) to plan and optimize the path, and the interaction layer provides intuitive navigation and user feedback mechanism. The system effectively solves the limitations of traditional guidance systems in complex environments and improves navigation efficiency and user experience. In terms of path planning efficiency, the average path planning time of the intelligent guidance system is shorter than that of the traditional system; in terms of path navigation accuracy, the average accuracy of the intelligent guidance system reaches 99.1%, which is much higher than the 95.2% of the traditional system. These data fully prove the effectiveness of the intelligent guidance system proposed in this paper in improving the quality of navigation services and user experience.

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1. Introduction

Traditional wayfinding system design methods mainly rely on static signs and manual planning, which makes it difficult to respond to complex and changing environments and diverse user needs in real time. In large public places with dense traffic, such as airports, shopping malls, hospitals, etc., traditional wayfinding systems often lead to poor user experience and even affect public safety due to problems such as information lag and inflexible path planning.

This paper introduces an intelligent navigation system that integrates DQN and YOLO. By introducing DQN, the system autonomously learns and optimizes path planning strategies to adapt to complex and changing environments; at the same time, combined with YOLO, it realizes real-time perception of the environment and information recognition, improving the accuracy and efficiency of navigation. The intelligent navigation system designed in this paper adopts a multi-level architecture, realizing a complete process from environmental perception to path planning, and then to user interaction.

The paper first outlines the research status and existing problems in the field of wayfinding system design, and introduces the research motivation and purpose of this paper. It introduces the design method of the intelligent wayfinding system, including the design and implementation of system architecture design, perception layer, data processing layer, decision layer and interaction layer. Then verifies the performance of the system through experiments, compares it with traditional systems, and demonstrates its advantages. Finally, it discusses the application prospects and future development directions of the system.

2. Related Work

As an important part of urban space and public facilities, the design of the sign system not only carries the function of information transmission, but also plays a key role in improving user experience, shaping the city image and promoting cultural inheritance. Xu et al. [1] combined the theory of design semiotics and analyzed the design method of the Sichuan West Forest Park sign system from the four dimensions of semantics, pragmatics, context and language structure of element symbols. They deconstructed the symbolic system of the sign system with modeling elements, colors, material textures and other objects; at the same time, they combined it with the four-dimensional system of "environment-scenery-people-objects" to construct the Sichuan West Forest Park sign system. Fang et al. [2] took Wuhan Metro as an example, and evaluated the effectiveness of the current sign system and identified the pain points through qualitative and quantitative methods. Based on the model, they classified and analyzed the pain points of the subway sign system from three levels: perception processing, memory storage, and decision execution. Finally, they constructed a subway sign information processing model and proposed three strategies: performance layer design based on perception, structural layer design based on memory, and behavior layer design based on decision-making. Wang Hetao[3] explored the cultural connotation of Shaanxi History Museum, sorted out and summarized the current status of the construction of the Shaanxi History Museum's guide system and user experience, explored the design methods, practical process and ideas of the Shaanxi History Museum's guide system, and expounded the theoretical research results of the guide system design to provide guidance for the design of the Shaanxi History Museum's guide system. Ding et al. [4] analyzed the actual application of high-speed rail station signage systems in transfer hubs, closely followed the development trends and latest developments of urban smart transportation signage systems, and worked hard to solve core problems such as visual distress and insufficient guidance that users may encounter during transfers, and proposed targeted optimization design solutions on this basis. Based on the fact that signage system design is limited to traditional methods and means, and is confined to a single discipline, resulting in a serious phenomenon of homogeneity in thousands of cities, and a lack of diversified integration and penetration, Lu and Zhan [5] reconstructed the "grand design" concept of signage systems, providing new ideas and research references for current signage system design research.

From the above research, we can see that the field of guide system design has achieved a series of important results, but it also faces the limitations of traditional design methods, especially when dealing with complex environments and diverse user needs. Traditional methods often seem to be unable to cope with the situation. At the same time, the successful application of reinforcement learning technology in multiple fields has demonstrated its powerful decision-making optimization capabilities and adaptability, providing new ideas and methods for solving complex problems in guide system design. Raffin et al. [6] implemented a variety of model-free deep reinforcement

learning algorithms, following software engineering best practices, and the code was benchmarked and automated unit tested. Brunke et al. [7] reviewed the transition from learning-based control to safe reinforcement learning and explored how to achieve safe learning of robots through reinforcement learning. Gronauer and Diepold[8] reviewed multi-agent deep reinforcement learning and analyzed reinforcement learning algorithms and architectures in multi-agent environments. Wurman et al.[9] used deep reinforcement learning algorithms, combined with advanced training techniques and model architectures, to develop AI that can surpass top human players in games. Rolf et al.[10] reviewed reinforcement learning algorithms and their applications in supply chain management, analyzing the advantages and disadvantages of different algorithms and their applicable scenarios. In the process of continuous exploration and innovation in the field of guide system design, people gradually realize the limitations of traditional design methods, and the successful application of reinforcement learning technology in many fields provides new ideas and methods to solve this problem. This paper introduces reinforcement learning and computer vision algorithms to break the limitations of traditional guide system design and explore a more intelligent, dynamic and personalized guide system design method. By combining the decision-making optimization capabilities of reinforcement learning with the environmental perception and information recognition capabilities of computer vision, this paper will strive to build an intelligent guidance system that can respond to user needs in real time and adapt to complex environmental changes, providing users with a more efficient and accurate navigation experience, while also injecting new vitality and innovative directions into the field of guidance system design.

3. Methods

3.1. System Architecture Design

The intelligent guidance system designed in this paper adopts a multi-level architecture, which mainly includes the perception layer, data processing layer, decision layer and interaction layer. The perception layer collects environmental information such as crowd density and obstacle location in real time by deploying cameras and sensor networks, and transmits these data to the data processing layer after preliminary preprocessing. The data processing layer uses YOLO to perform operations such as target detection and semantic segmentation on the data transmitted from the perception layer, extract environmental features and build a dynamic environmental model to provide real-time environmental information for the decision layer.

The decision layer uses the deep Q network (DQN) reinforcement learning algorithm to construct the state space based on the environmental model provided by the data processing layer, combined with key information such as user location, target location, obstacle distribution, etc., and defines navigation actions such as moving forward, turning, and waiting as the action space [11-12]. At the same time, the model is guided to perform path planning and decision optimization by designing a reward function (taking into account factors such as path length, time consumption, path safety and comfort). The decision layer undergoes extensive training in a simulated environment, enabling it to dynamically adjust the path planning strategy based on real-time environmental information, generate the optimal navigation path, and feed the results back to the interaction layer.

The interactive layer develops a navigation interface based on mobile terminals to provide users with intuitive navigation instructions, including arrow instructions, voice prompts, etc. At the same time, the interactive layer also has a user feedback mechanism to allow users to feedback on their navigation experience. The system architecture is shown in the figure 1:

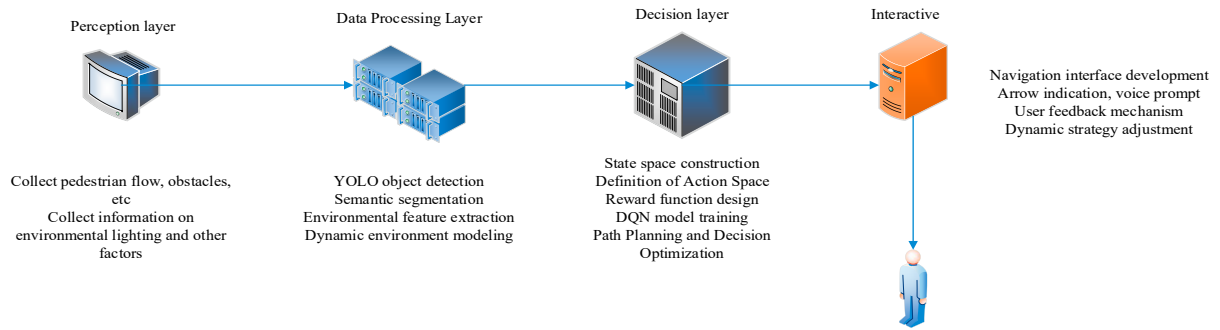


Fig 1. Guide system architecture

Through this multi-level architecture design, each level works closely together to achieve a complete process from environmental perception to path planning to user interaction, ensuring the efficiency, accuracy and personalization of the intelligent guidance system.

3.2. Perception Layer

In the intelligent guidance system, the perception layer realizes the real-time collection of environmental data such as crowd density and obstacle location through the deployment of cameras and sensor networks. The perception layer obtains environmental information by deploying cameras and sensor networks. As an important perception device, the camera captures scene images in real time for crowd density estimation and obstacle detection. In addition, the sensor network (infrared sensor, ultrasonic sensor, etc.) is used to detect the location of obstacles and the movement trajectory of people. The collected data is shown in Table 1:

Table 1. Data collected by the perception layer

Timestamp	Area	People Density (people/m ²)	Obstacle Location (X, Y Coordinates)	Obstacle Type
2023-10-01 08:00:00	Area A	0.6	(50, 100)	Stationary Obstacle
2023-10-01 08:01:00	Area B	1.2	(150, 200)	Moving Obstacle
2023-10-01 08:02:00	Area A	0.3	(50, 100)	Stationary Obstacle
2023-10-01 08:03:00	Area C	0.7	(250, 300)	Stationary Obstacle
2023-10-01 08:04:00	Area D	1.0	(350, 400)	Moving Obstacle
2023-10-01 08:05:00	Area B	0.8	(100, 150)	Stationary Obstacle
2023-10-01 08:06:00	Area A	0.5	(200, 250)	Moving Obstacle
2023-10-01 08:07:00	Area C	0.9	(300, 350)	Stationary Obstacle

These devices work together to provide the system with comprehensive environmental perception capabilities. Through cameras and AI algorithms, the system can accurately calculate parameters such as crowd flow, density, and walking speed. At the same time, the sensor network can detect the location of obstacles, help the system update environmental information in real time, and ensure the accuracy of the navigation path. Through the deployment of the perception layer and data collection, the guidance system can obtain environmental information in real time, providing basic data support for subsequent data processing and path planning.

3.3. Data Processing Layer

The data processing layer applies the YOLO algorithm and semantic segmentation (DeepLab) algorithm, and combines data preprocessing and feature fusion technology to ensure the accuracy and real-time nature of environmental information.

The YOLO algorithm can accurately identify multiple target objects in an image in a short time. In the intelligent guidance system, the YOLO algorithm is used for target detection and environmental feature extraction [13-14]. It

quickly detects key target objects such as pedestrians, obstacles, and signs in the image and extracts the target's feature vector. These feature vectors not only contain the location and category information of the target but can also be used to further analyze the target's motion trend, density distribution, etc.:

$$D = \{(x_i, y_i, w_i, h_i, c_i)\}_{i=1}^N \quad (1)$$

D is the algorithm output, (x_i, y_i) refers to the center coordinate of the target, w_i and h_i refer to the width and height of the target, respectively, c_i refers to the category of the target, N is the number of detected targets, and i is its index.

In the system, the DeepLab semantic segmentation algorithm is used for environmental semantic segmentation and dynamic environment modeling. By performing pixel-level semantic segmentation on the images transmitted by the perception layer, DeepLab classifies each pixel in the image into different semantic categories, such as segmenting roads, pedestrians, obstacles, signboards, etc., to form an environmental semantic map. Based on the results of semantic segmentation, the system builds a dynamic environment model to reflect the position, shape, and category information of each element in the environment in real time, providing an accurate description of the environment for the decision-making layer. The output of semantic segmentation is represented as a semantic label matrix S :

$$S = \{s_{ij}\}_{i,j=1}^{H,W} \quad (2)$$

s_{ij} refers to the semantic category of the pixel in the i -th row and j -th column of the image, and H and W are the height and width of the image, respectively.

Data preprocessing mainly includes image denoising, data normalization and resizing. Image denoising is used to remove noise interference in the image data transmitted from the perception layer to improve image quality; data normalization normalizes the image data to a uniform range of 0 to 1; and resizing adjusts the image to a suitable size according to the algorithm requirements.

3.4. Decision-Making Level

In the decision-making layer, this study uses DQN to dynamically adjust the navigation path by learning the optimal strategy to ensure that the system can guide users to the target location efficiently and accurately [15].

The state space is the basis of the DQN algorithm. It describes all the key information that the intelligent guidance system needs to consider in the decision-making process. In this system, the state space mainly includes the user position, target position, obstacle distribution, and environmental characteristics. These state information together constitute the state space of the intelligent guidance system, providing a comprehensive environmental description for the DQN algorithm, enabling it to make accurate decisions:

$$S = \{u, g, o, f\} \quad (3)$$

S is the state space, u is the current position of the user in the environment, g is the destination the user expects to reach, o is the location and distribution of obstacles in the environment, and f is the environmental characteristics including crowd density, path connectivity, etc.

The action space defines the possible actions that the intelligent guidance system can take in each state. In this system, the action space mainly includes navigation actions such as moving forward, turning, and waiting; moving forward means continuing in the current direction; turning means adjusting the direction of travel according to the target location and obstacle distribution; waiting means pausing in certain situations. These actions are selected and optimized by the DQN algorithm to achieve the optimal navigation path planning:

$$A = \{a_{\text{forward}}, a_{\text{turn}}, a_{\text{wait}}\} \quad (4)$$

The action space A in formula (4) mainly includes navigation actions such as forward a_{forward} , turn a_{turn} and wait a_{wait} .

The reward function comprehensively considers factors such as path length, time consumption, safety, and comfort. Path length encourages the system to choose a shorter path to reach the target location; time consumption optimizes navigation time and reduces the total time it takes for users to reach the target; safety prevents users from colliding with obstacles and ensures the safety of the navigation process; comfort considers the user's experience during navigation and avoids frequent turns or sudden stops. By reasonably designing the reward function, the DQN algorithm learns the optimal navigation strategy and balances the relationship between path length, time, safety, and comfort.

3.5. Interaction Layer

The interactive layer is the part of the intelligent guidance system that comes into direct contact with users. It provides users with an intuitive and convenient navigation experience, and dynamically adjusts the navigation path based on user feedback to ensure personalized and real-time services.

The interactive layer is mainly realized through the navigation interface based on mobile terminals. Mobile terminals have become the main carrier of navigation interfaces due to their portability and popularity. Users obtain real-time navigation information through applications on these devices. The system provides users with clear navigation instructions through arrow instructions, voice prompts and other methods. During navigation, the user interface updates the path information in real time to respond to changes in the environment or the user's location. When the user approaches an intersection, the system will clearly inform the user of the direction to turn through arrow instructions; at the same time, the voice prompt function will issue voice instructions to remind the user of information such as "turn left 5 meters ahead". This multi-modal interaction method not only improves the accuracy of navigation but also enhances the convenience and comfort of the user experience.

The user feedback mechanism allows users to provide feedback on their experience during navigation. Users can use the feedback button on the navigation interface to report whether the path is smooth, whether there are obstacles that have not been detected, and other issues.

4. Results and Discussion

4.1. System Performance Test

In the process of exploring the performance boundaries of the intelligent guidance system, direct comparison with the traditional static guidance system is the key to verifying its innovative value. To this end, this paper designs and implements a series of comparative experiments, aiming to use quantitative indicators to outline the performance of the intelligent guidance system that integrates YOLO and DQN in path planning efficiency and path navigation accuracy.

The experiment simulates a variety of complex navigation scenarios, including dense crowds, variable obstacle settings, etc. Subsequently, the same starting and ending information is input into the traditional static guidance system and the intelligent guidance system to simulate the user's actual navigation needs. During the experiment, the system records the data of the two systems in terms of path planning efficiency and path navigation accuracy. The results are shown in Figures 2 and 3, respectively:

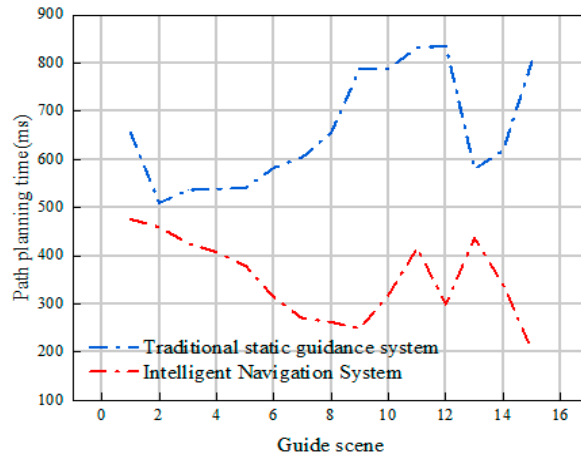


Fig 2. Path planning time

As shown in Figure 2, from the overall trend, the path planning time of the intelligent guidance system in most scenarios is shorter than that of the traditional guidance system. This shows that the intelligent guidance system that integrates reinforcement learning and computer vision algorithms has higher efficiency in handling complex navigation tasks. In scenarios 1 to 5, although the path planning time difference between the two is not large, the intelligent guidance system still shows certain advantages. This is because these scenarios are relatively simple and the traditional guidance system can also generate navigation paths quickly. However, as the complexity of the scenes increases, the advantages of the intelligent guidance system gradually become apparent. Starting from scene 6, the path planning time of the intelligent guidance system is significantly shorter than that of the traditional guidance system. In particular, in scenes 9 and 10, the intelligent guidance system can quickly adapt to different navigation needs and complete the path planning in 249 milliseconds and 319 milliseconds, respectively. In contrast, traditional guidance systems take longer to process these changes. This efficiency improvement is not only due to the advanced algorithm architecture of the intelligent guidance system but also to its ability to analyze and understand environmental data in real time, thereby making more accurate decisions.

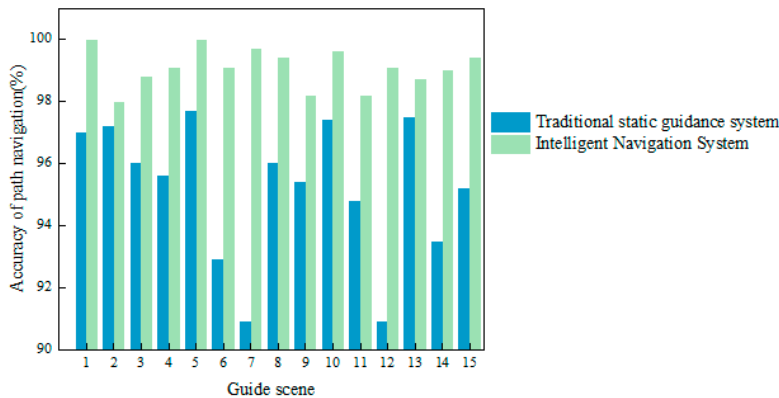


Fig 3. Route navigation accuracy

As shown in Figure 3, the average accuracy of the path navigation of the traditional guidance system is about 95.2%, while the average accuracy of the intelligent guidance system is about 99.1%. This gap shows that the intelligent guidance system has improved the accuracy of navigation by integrating reinforcement learning and

computer vision algorithms. The high average accuracy of the intelligent guidance system not only reflects its superiority in handling complex navigation tasks but also reflects its stability and consistency in different scenarios. No matter what kind of navigation challenge it faces, the intelligent guidance system can maintain a high level of navigation accuracy and provide users with more reliable and accurate navigation services. The intelligent guidance system can dynamically adjust navigation strategies by analyzing and understanding environmental data in real time to ensure navigation accuracy and efficiency.

4.2. Application Prospects

In large public places such as airports, shopping malls and hospitals, the intelligent guidance system can not only quickly and accurately guide users to their destinations, but also adjust the route planning according to real-time conditions to optimize the user experience. In terms of smart city construction, it will be deeply integrated with intelligent transportation systems, tourist attraction management and public transportation hubs to improve urban operation efficiency and residents' quality of life.

The industrial and logistics sectors also benefit from the application of intelligent guidance systems. In smart warehousing, factory workshops and logistics distribution, the system provides efficient cargo positioning, route planning and distribution services, reducing costs and improving efficiency. Educational and cultural venues, such as libraries, museums and campuses, will also achieve more convenient navigation and information push services through intelligent guidance systems to meet the diverse needs of users.

In addition, the intelligent guidance system also has application value in special scenarios, such as emergency evacuation and barrier-free navigation, contributing to public safety and social inclusiveness. With the continuous advancement of technology, multimodal data fusion, cross-scenario adaptability, personalized service upgrades, and deep integration with smart cities will become important development directions for intelligent guidance systems. By optimizing algorithms and hardware design, the system will further reduce power consumption and improve real-time performance, ensuring stable and reliable navigation services in various environments.

5. Conclusion

This paper addresses the limitations of traditional guidance systems when facing complex environments and diverse user needs. The intelligent guidance system designed in this paper adopts a multi-level architecture, realizing the complete process from environmental perception to path planning and then to user interaction. Compared with traditional static guidance systems, the intelligent guidance system has advantages in both path planning efficiency and path navigation accuracy. Although the system design method proposed in this paper has solved the limitations of the traditional guide system to a certain extent, the personalized service capabilities of the system need to be further improved to better meet the specific needs of different users. In the future, we will further explore directions such as multi-data fusion, cross-scenario adaptability, and personalized service upgrades to improve the intelligence level of the system and user experience. At the same time, combined with the development trends in the fields of smart cities and intelligent transportation, it will play a more important role in urban space and public facilities, bringing greater convenience and comfort to people's lives.

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Project Title: Research on the Path of Data-Enabled Personalized Cultivation of Students in the Perspective of Industry-Education Integration

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